

Segment-1

Evolution of Microprocessor System Bus Hardware of Microprocessor

Microprocessor, microcontroller and embedded system

Course code: CSE-3523

Credit Hour: 3

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Old Syllabus

1	Introduction: Introductory Concept: Evaluation of microprocessor, Types of microprocessor, system bus, hardware of a microprocessor, memory-addressing technique.	02	CLO1
2	8086 Microprocessor: properties, architecture, registers, FLAGS register, physical address calculation, addressing modes, Addressing Techniques of 8086 Microprocessor.	06	CLO1
3	Instruction: Instruction set, Instruction format, Fetch-decode-Execution cycle, basic concepts of assembly language	04	CLO2
Section-B (SEE: 50 Marks)			
4	Interrupt System: Sources of interrupt, Types of interrupt, handling interrupt request, interrupt vector and table, 8259A priority interrupt controller, Daisy chain..	04	CLO1
5	An overview of Intel 80186, 80286, 80386, 80486 and Pentium microprocessor; Advanced microprocessor: Embedded microprocessor, Bit slice microprocessor, arithmetic processor, Multitasking, Itanium and Merced Microprocessor.	06	CLO1
6	I/O operation: Isolated and memory mapped I/O, 8255A Programmable peripheral Interface, DMA technique, I/O ports, I/O processor	02	CLO1
7	Micro controller: Architecture of 8051, Signals, Operational features, Memory and I/O addressing, Interrupts, Instruction set, Applications. Design of a particular microcontroller based architecture.	04	CLO3
8	Embedded Systems: Introduction to Embedded Systems, Characteristics, Components, Hardware-Software Partitioning, Specification and Modeling, Optimizing Design, SOC, and Application Domains.	02	CLO1
Course Teacher: Mujibur Rahman Maruf		30	

New Syllabus:effective from Autumn 25

#	Content	Duration	CLOs
Mid-Term (30 Marks)			
1	Introductory Concept: Evaluation of microprocessor, Types of microprocessor, system bus, hardware of a microprocessor, memory-addressing technique.An overview of Intel 80186, 80286, 80386, 80486 and Pentium microprocessor;	2 weeks	CLO1
2	8086 Microprocessor: properties, architecture, registers, FLAGS register, physical address calculation, addressing modes, Addressing Techniques of 8086 Microprocessor.	2 weeks	CLO1, CLO2
3	Instruction: Instruction set, Instruction format, Fetch-decode-Execution cycle.	2 weeks	CLO2

Final Exam: 50 Marks			
Group-A (20 Marks)			
4	Interrupt System: Sources of interrupt, Types of interrupt, handling interrupt request, interrupt vector and table, 8259A priority interrupt controller, Daisy chain.	2 weeks	CLO2
5	Advanced microprocessor: Embedded microprocessor, Bit slice microprocessor, arithmetic processor, Multitasking, Itanium and Merced Microprocessor, Pipelining,DMA technique,	2 weeks	CLO2
Group-B (30 Marks)			
6	I/O operation: Isolated and memory mapped I/O,Basic Input and Output Interfaces,Handshaking,Address decoding ,8255A Programmable peripheral Interface, I/O ports, I/O processor	1 week	CLO3
7	Microcontroller and Embedded Systems: Architecture of 8051, Signals, Operational features, Memory and I/O addressing, Interrupts, Instruction set, Applications.Introduction to Embedded Systems, Characteristics, Components, Hardware-Software Partitioning, Specification and Modeling, Optimizing Design, SOC, and Application Domains.	2 weeks	CLO3
8	IoT and its Technology: The Internet of Things basics, Infrastructure, Networks and Communication, Processes, Data Management, Trust M2M to IoT, M2M Value Chains, IoT Value Chains, IoT Architecture—State of the Art, Value Creation from Big Data and Serialization, Applications of IOT, privacy and security an overview towards Governance policy, First Steps Towards a Secure Platform, Smart Approach, Data Aggregation for the IoT in Smart Cities, and Security.	2 weeks	CLO3

Activate
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Reference books

1. Barry B. Brey, **The Intel Microprocessors**, 17th ed., Prentice Hall of India (PHI) Pvt. Ltd.
2. Douglas V Hall, **Microprocessors and Interfacing**, 2nd ed., Macmillan/McGraw Hill.
3. MD. Rafiquzzaman, **Microprocessor and Microcomputer based system design**, 2nd ed.

A COMPUTER SYSTEM

- A computer is a programmable machine that receives input, stores and manipulates data//information, and provides output in a useful format.



Diagram Of A Computer System

16

The Microprocessor

- ❑ Microprocessor, the heart of the microprocessor-based computer system is an integrated circuit (IC).
- ❑ Microprocessor is the **controlling element** in a computer system.
- ❑ It **controls memory and I/O devices** through buses and **transfer instructions and data** between microprocessor and memory or I/O device.



Main Tasks of Microprocessor

- ❑ The microprocessor performs three main tasks for the computer system:
 1. **Data transfer** between microprocessor and memory or I/O devices.
 2. Simple **arithmetic and logic operations**.
 3. **Program flow** via simple decisions.
- ❑ The power of the microprocessor is in its capability to execute billions of millions of instructions per second from a program or software (group of instructions) stored in memory.

Evolution of The Microprocessor

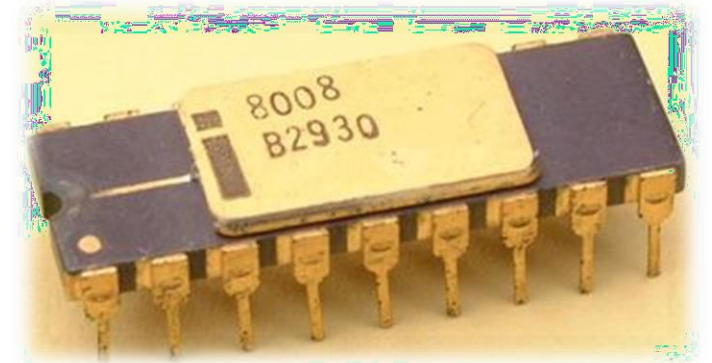
❑ 1st Microprocessor

- ✓ 1971, INTEL (INTEgrated ELectronics), 4004/4040
- ✓ 4-bit microprocessor
- ✓ 4-bit memory, contain only 45 instructions
- ✓ Fabricated with P-channel MOSFET
- ✓ Execute instructions at slow rate of 50 KIPs (Kilo-instructions per second)



❑ 2nd Microprocessor

- ✓ Later in 1971, INTEL corp., 8008
- ✓ 8-bit microprocessor
- ✓ 16kB memory, contain only 48 instructions,
- ✓ Execution time 20 micro-seconds



Evolution of The Microprocessor

❑ 3rd Microprocessor (**1st modern microprocessor**)

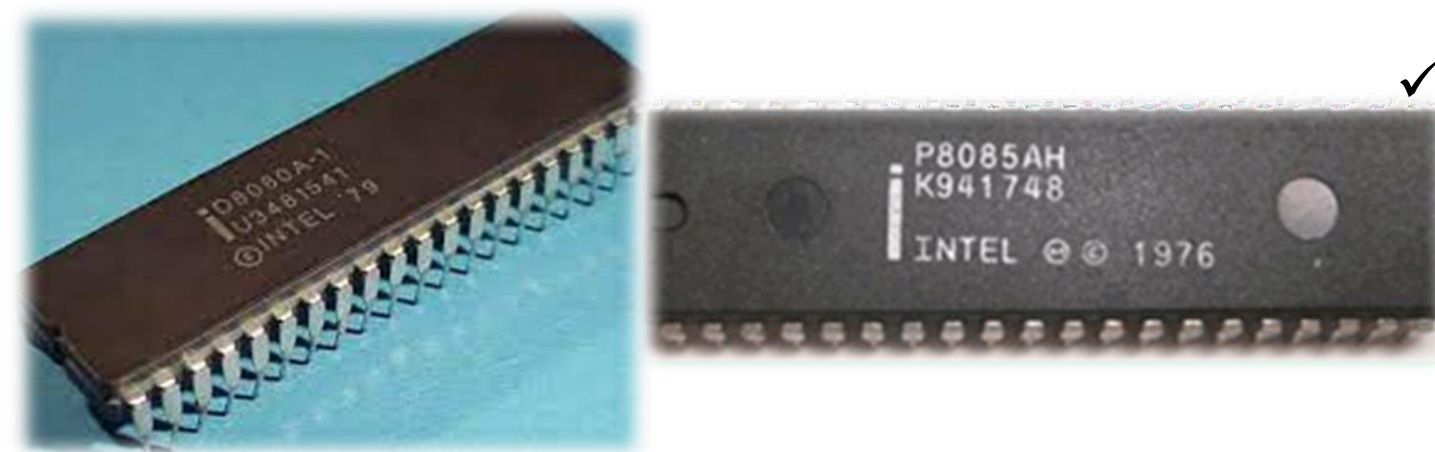
- ✓ 1973, INTEL 8080
- ✓ 8-bit microprocessor
- ✓ 64kB memory
- ✓ TTL technology
- ✓ Execution time 2 micro-seconds

❑ The 8085 Microprocessor

- ✓ 1977, Intel 8085
- ✓ 8-bit microprocessor (Data bus 8 bits, address bus 8 bits)
- ✓ 64kB memory
- ✓ Execution time 1.3 micro-seconds

Advantage:

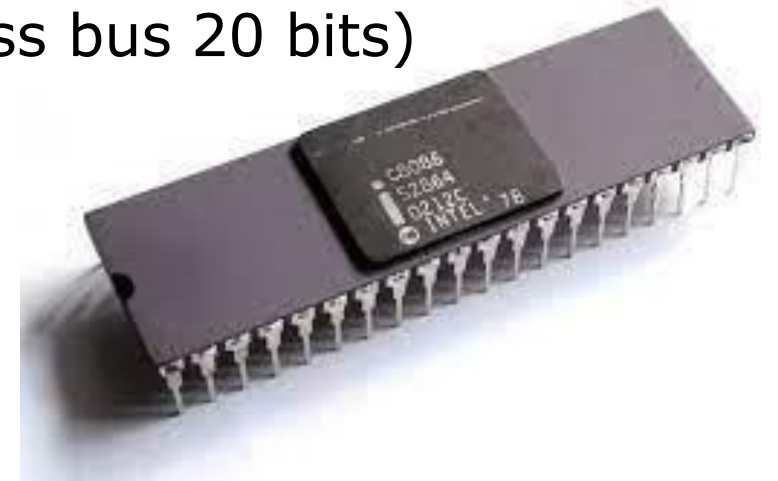
1. internal clock generator
2. Internal system controller
3. Higher clock frequency (3MHz)



Evolution of The Microprocessor

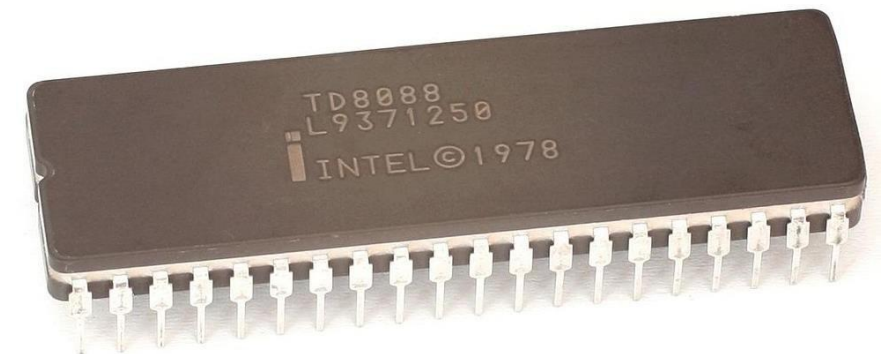
❑ The 8086 Microprocessor

- ✓ 1978, Intel 8086
- ✓ 16-bit microprocessor (Data bus 16 bits, address bus 20 bits)
- ✓ 1MB memory
- ✓ Execution time 400 nano-seconds
- ✓ Clock frequency 5MHz
- ✓ Cache or queue memory 6 byte



❑ The 8088 Microprocessor

- ✓ 1979, Intel 8088
- ✓ Same as 8086 but speedy than 8086



Evolution of The Microprocessor

❑ The 80286 Microprocessor

- ✓ 1983, INTEL 80286
- ✓ 16-bit microprocessor
- ✓ 16MB memory
- ✓ Execution time 250 nano-seconds
- ✓ Clock frequency 8MHz



❑ The 80386 Microprocessor (**1st practical 32-bit MP**)

- ✓ 1986, Intel 80386
- ✓ 32-bit microprocessor (Data bus 32 bits, address bus 32 bits)
- ✓ 4GB memory
- ✓ Cache or queue memory 8kB
- ✓ Have few modified version (80386SX, 80386SL/ 80386SLC, 80386EX)

Evolution of The Microprocessor

- ❑ The 80486 Microprocessor
 - ✓ 1989, INTEL 80486
 - ✓ Same as 80386
 - ✓ Execution time 25 nano-seconds
 - ✓ Clock frequency 50MHz



Evolution of The Microprocessor

❑ The Intel Pentium Microprocessor

- ✓ 1993
- ✓ 64-bit microprocessor
- ✓ 4GB memory
- ✓ Cache or queue memory 16kB
- ✓ Pentium contained 8KB instruction cache & 8KB data cache
- ✓ Execute instructions at the rate of 110 MIPS
- ✓ Two version of Pentium operated with a clocking frequency of 60MHz & 66MHz
- ✓ Double-clock Pentium, operating at 120MHz & 133MHz also available



Evolution of The Microprocessor

64 bit MP

Name	Year of Invention	Clock speed (GHz)	Number of transistors	Inst. per sec
INTEL core 2	2006 (other versions core2 duo, core2 quad, core2 extreme)	1.2 to 3	291 Million transistors	64 KB of L1 cache per core, 4 MB of L2 cache
i3, i5, i7	2007, 2009, 2010	2.2–3.3, 2.4–3.6, 2.93–3.33		

❑ Fifth-generation of MP

- From 1995 till now we are in the fifth generation. 64-bit processors like PENTIUM, Celeron, dual, quad, and octa-core processors came into existence.

Microcontroller

- ❑ A device/microcomputer that includes microprocessor, memory, several signal lines on a single chip fabricated using VLSI technology. A single chip microcomputer is called a microcontroller.
- ❑ The microcontroller on single chip, which typically includes a microprocessor, 64 byte RAM, 1-2 kilo byte ROM and several signal lines to connect I/O devices.
- ❑ Microcontroller are used for such as controlling appliances, traffic lights etc.
- ❑ Typical IC's example are Zilog-Z8, Intel MC551, Motorola 68HC11.

Microprocessor vs MicroController

Microprocessor	Microcontroller
Microprocessor is the heart of Computer system.	Micro Controller is the heart of an embedded system.
It is only a processor, so memory and I/O components need to be connected externally	Micro Controller has a processor along with internal memory and I/O components.
Memory and I/O has to be connected externally, so the circuit becomes large.	Memory and I/O are already present, and the internal circuit is small.
You can't use it in compact systems	You can use it in compact systems.
Cost of the entire system is high	Cost of the entire system is low

Some Important Terms

- ❑ Bit (b) - a binary digit, 0 or 1.
- ❑ Byte (B) - a group of 08 bits. (1 byte = 8 bits)
- ❑ Word (W) - a group of 02 bytes. (1 word = 2 bytes = 16 bits)
- ❑ Instruction - a command in binary, recognized & executed by the computer.

Microprocessor-based Computer System

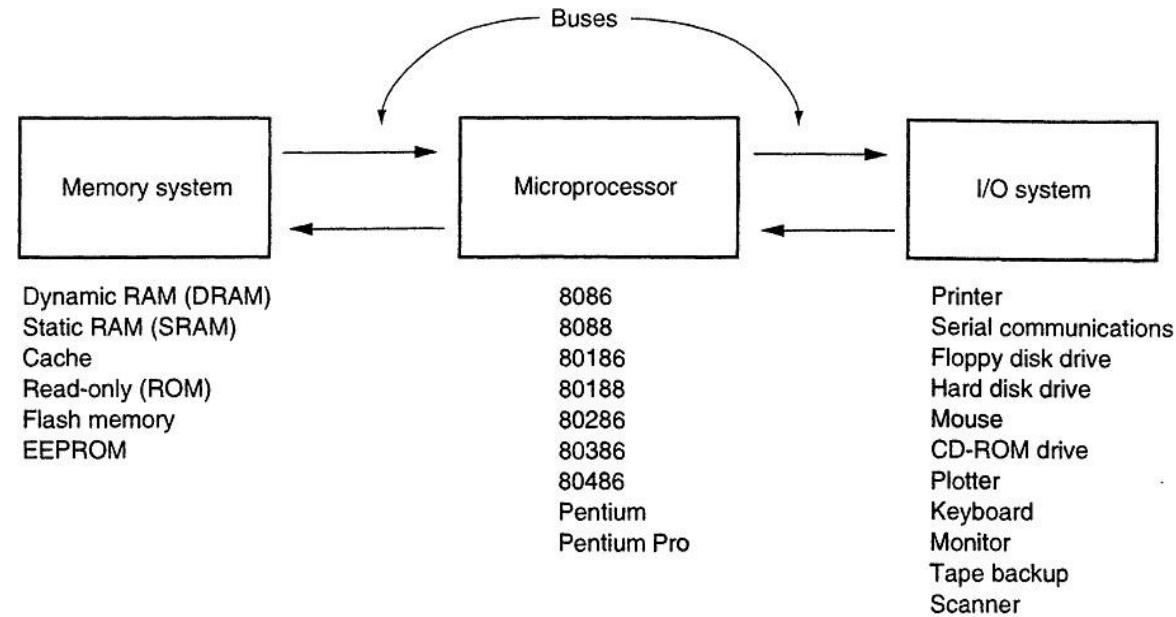


FIGURE 1-2 The block diagram of a microprocessor-based computer system

Types of Microprocessor

1. Complex Instruction Set Computer (CISC)

CISC or Complex Instruction Set Computer is a computer architecture where instructions are such that a single instruction can execute multiple low-level operations like loading from memory, storing into memory, or an arithmetic operation, etc.

Example: 1. Intel 386 2. Intel 486 3. Pentium

ADD A, B

This single instruction might:

- load A from memory
- add B to it
- store the result back in memory

Types of Microprocessor

2.Reduced Instruction Set Computer (RISC) is a processor design approach where the CPU uses a **small set of simple instructions**. Each instruction is designed to execute very quickly, usually within one clock cycle.

Instead of using complex instructions, RISC processors break tasks into **many small, simple steps**. Because the instructions are simple and uniform, the processor can execute them faster and more efficiently.

Instead of one complex instruction, the task is divided into steps:

LOAD R1, A

LOAD R2, B

ADD R3, R1, R2

STORE R3, C

Types of Microprocessor

3. Explicitly Parallel Instruction Computing (EPIC)

EPIC or Explicitly Parallel Instruction Computing permits computers to execute instructions parallel using compilers. It allows complex instructions execution without using higher clock frequencies.

In EPIC architecture:

- The compiler analyzes the program.
- It groups independent instructions together.
- These grouped instructions are executed at the same time by the CPU.

Suppose a program has these independent operations:

- $A + B$
- $C + D$
- $E + F$

In EPIC, the compiler may group them like this:

`{ ADD A, B } { ADD C, D } { ADD E, F }`

These can run in parallel during the same cycle if hardware units are available.

Types of Microprocessor

4. Superscalar microprocessor

A **Superscalar Microprocessor** is a CPU design that can **execute more than one instruction in a single clock cycle**. It does this by having multiple execution units inside the processor, allowing several instructions to ☐ run at the same time.

If there are independent instructions like:

ADD R1, R2, R3

SUB R4, R5, R6

MUL R7, R8, R9

A superscalar CPU may execute two or more of these **at the same time** in one clock cycle.

5. Application-Specific Integrated Circuit (ASIC) is a type of integrated circuit that is **designed for a specific application or task**, rather than for general-purpose use. ☐ These processors are application-specific like personal digital assistant computers. They are designed according to proper specifications.

ASICs are commonly used in:

- Smartphones (camera processing, communication chips)
- Network devices like routers and switches
- Cryptocurrency mining hardware
- Consumer electronics (TVs, washing machines, etc.)

Characteristics of a microprocessor

1. It assimilates the functions of a CPU on to a single IC.
2. It can perform some basic operations like addition, subtraction, multiplication, division and some logical operations using its Arithmetic and Logical Unit (ALU).
3. Move data from one location to another.
4. It has a program counter that can store the addresses of next instruction.
5. Computational capacity is very high.
6. Perform the instruction cycle like as fetch, decode and execute repeatedly.
7. To design a system with a microprocessor, it needs to connect with RAM, ROM, I/O devices etc.
8. Building cost is high.
9. Power consumption and dissipation is high.

Characteristics of a microprocessor

Three basic characteristics that differentiate microprocessors are-

- **Instruction set:** The set of instructions that the microprocessor can execute. MIPS stands for "millions of instructions per second" and is a rough measure of the performance of a CPU.
- **Bandwidth or Data Width:** The number of bits processed in a single instruction. Data Width is the width of the ALU. An 8-bit ALU can add/subtract/multiply/etc. two 8-bit numbers, while a 32-bit ALU can manipulate 32-bit numbers.
- **Clock speed:** Given in megahertz (MHz), it determines how many instructions per second the processor can execute. For example, a 32 bit microprocessor that runs at 50MHz is more powerful than a 16-bit microprocessor that runs at 25MHz.

Hardware of a microprocessor

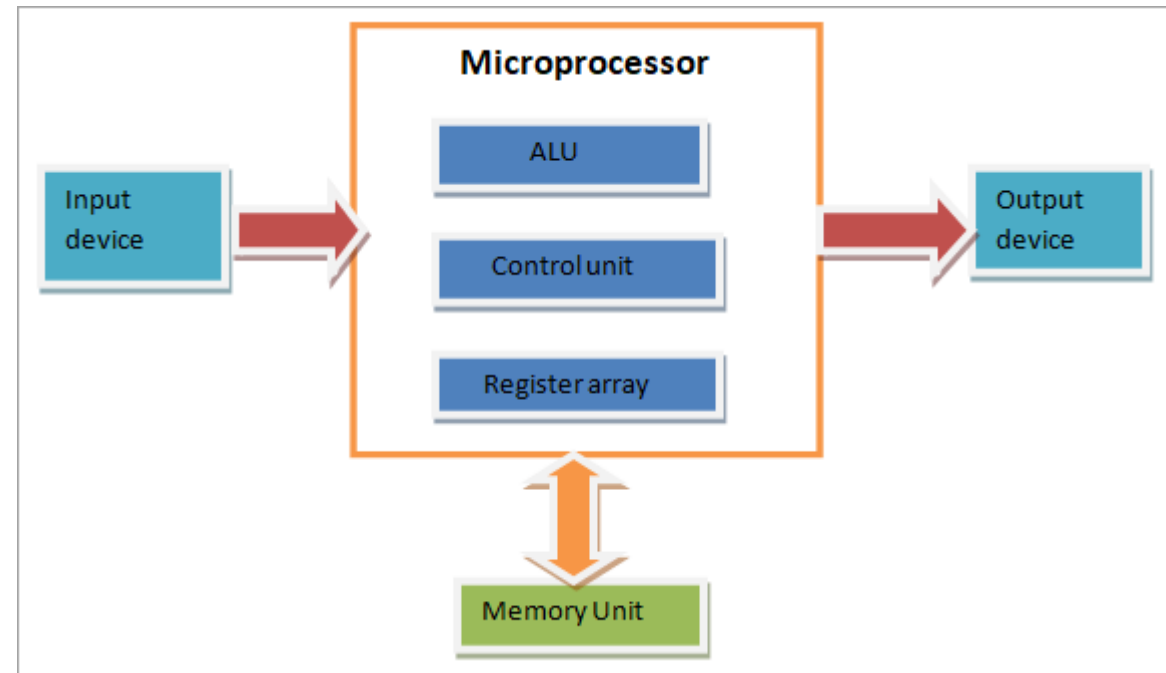
Arithmetic Logic Unit (ALU)

Performs arithmetic & logical operations.

Control Unit (CU) – Directs and manages instruction execution.

Registers – Small storage locations for quick data access.

Buses (Data, Address, Control) – Transfers data and signals between components



Central Processing Unit (CPU)

Central Processing Unit (CPU) – The Brain of the Computer

- The CPU controls and executes all computer tasks.
- It performs arithmetic & logical operations efficiently.

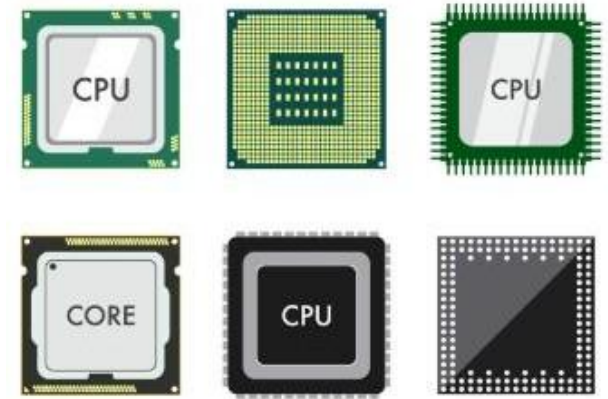
Main Components:

- ALU (Arithmetic Logic Unit): Handles calculations & logic.
- CU (Control Unit): Directs and manages operations.

How It Works?

1. Receives input from the user.
2. Processes data using ALU & CU.
3. Generates and delivers output.

The CPU is the heart of a computer, ensuring smooth and efficient functioning of all operations.



Arithmetic & Logic Unit (ALU)

i. Arithmetic & Logic Unit (ALU)

- Some of the tasks performed by the ALU are given below:

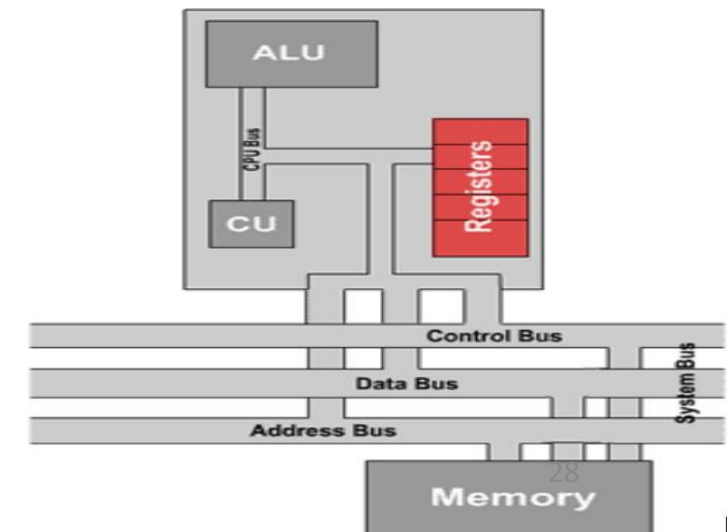
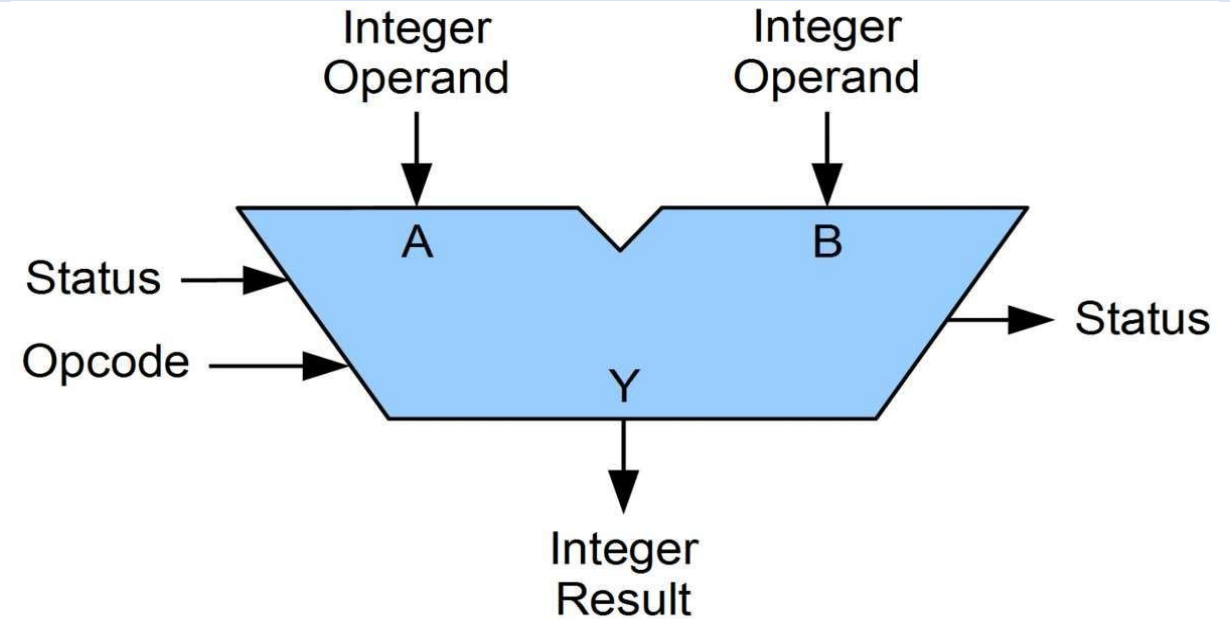
Addition and subtraction:.

Multiplication and division:

Logical tests:

Comparison:

Bit shifting:



Control Unit (CU)

❑ controls the movement of instructions in and out of the processor, and also controls the operation of the ALU..

The three main elements of the control unit

Decoder: This is used to decode the instructions that make up a program when they are being processed.

Timer or clock: The timer or clock ensures that all processes and instructions are carried out and completed at the right time

Control logic circuits: used to create the control signals themselves, which are then sent around the processor..

Memory Array Register

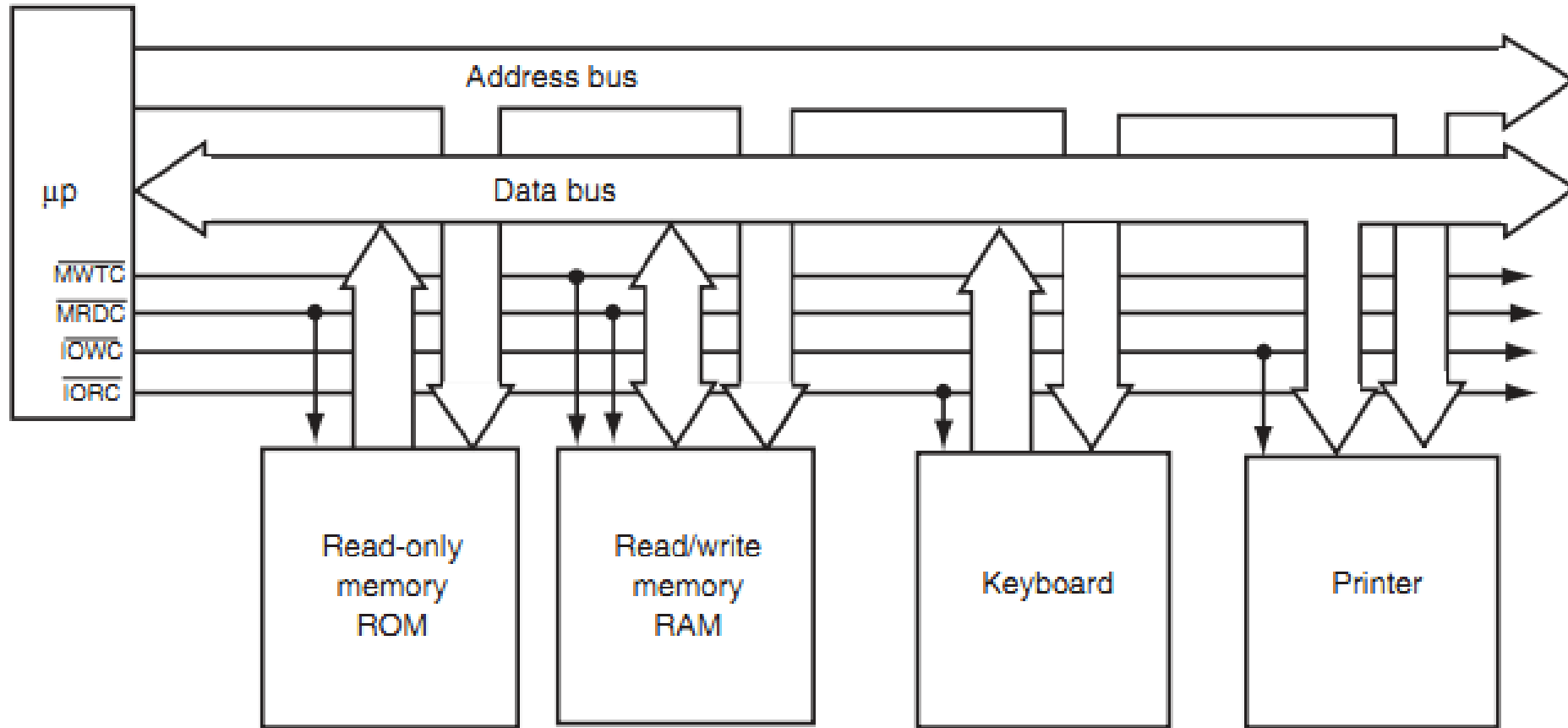
some common registers used for specific functions. These are:

- ❑ **Program Counter (PC):** used to hold the memory address of the next instruction that has to execute in a program.
- ❑ **Instruction Register (IR):** used to hold the current instruction in the processor while it is being decoded and executed.
- ❑ **Accumulator (A, or ACC):** used to hold the result of operations performed by the arithmetic and logic unit.
- ❑ **Memory Address Register (MAR):** Used for storage of memory addresses, usually the addresses involved in the instructions held in the instruction register

Memory Array Register

- ❑ **Memory Buffer Register (MBR):** When an instruction or data is obtained from the memory or elsewhere, it is first placed in the memory buffer register.
- ❑ **Flag register / status flags:** The flag register is specially designed to contain all the appropriate 1-bit status flags,
- ❑ **Other general purpose registers:** generally used for the quick storage of pieces of data that are required later in the program execution..

Buses:system Bus



Buses....

*Write - MP to others
Read - others to MP*

Block diagram of bus structure

Buses:system Buses

- ❑ A group of lines used to **transfer bits between microprocessor and other components** (microprocessor, RAM, ROM, I/O) of computer is called bus.
- ❑ There are 3 types of System buses-
 - 1. Data bus (Bi-directional)**
 - **Transfers information** between microprocessor and its memory and I/O address space.
 - 2. Address bus (Unidirectional)**
 - **Requests a memory location** from the memory or I/O location from I/O devices.

Buses:system Buses

3. Control bus (Unidirectional)

- Contains lines that **select memory or I/O** and cause them to **perform a read or write operation**.
- Four control bus connections:
 - MRDC – Memory read control
 - MWTC – Memory write control
 - IORC – I/O read control
 - IOWC – I/O write control

Function of Buses....

- **1. Data sharing** All types of buses found on a computer must be able to transfer data between the computer peripherals connected to it.
- **2. Addressing** -A bus has address lines, which match those of the processor. This allows data to be sent to or from specific memory locations.
- **3. Power** A bus supplies power to various peripherals that are connected to it.
- **4. Timing** The bus provides a system clock signal to synchronize the peripherals attached to it with the rest of the system.

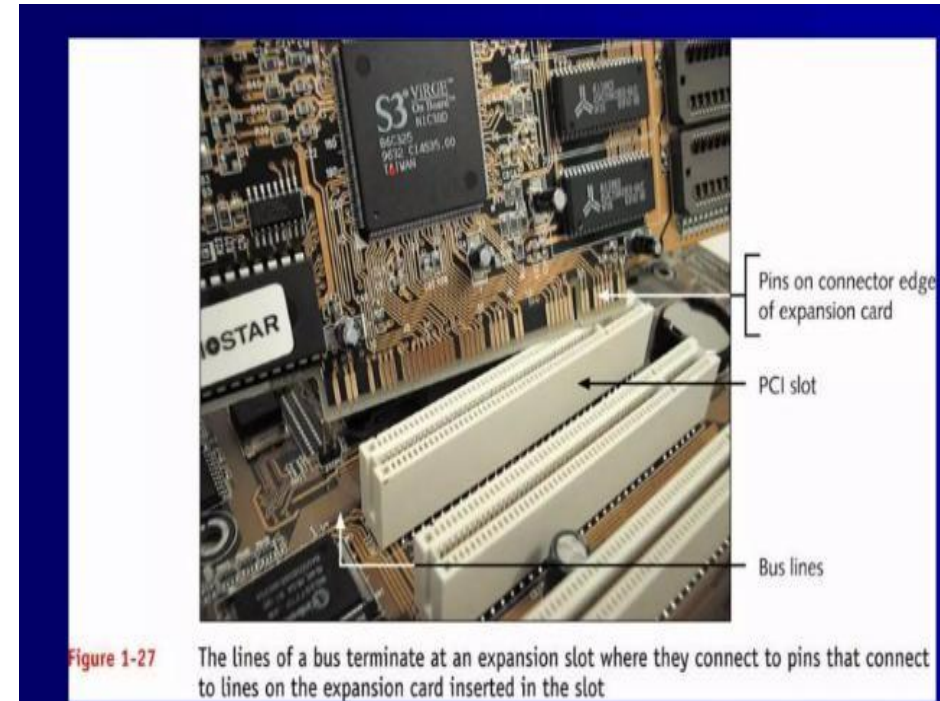
Expansion Type Buses....

Integrated Motherboards

- Common devices (keyboard, mouse, audio, video, LAN) are built-in.
- Lower overall cost, simpler design.
- Expensive to repair, limited customization.
- Not ideal for high-performance applications like gaming or advanced graphics.

Expansion Cards

- Used for specialized devices requiring higher performance.
- Installed in expansion slots on the motherboard.
- Examples: Graphics Cards, Sound Cards, Network Cards.



Expansion Type Buses....

These are some of the common expansion bus types that have ever been used in computers

- **PCI** - Peripheral Component Interconnect
- **PCMCIA** - Personal Computer Memory Card Industry Association (Also called PC bus)
- **AGP** - Accelerated Graphics Port
- **SCSI** - Small Computer Systems Interface

1. Peripheral Component Interconnect (PCI)

- is a hardware bus standard used to connect peripheral devices to the motherboard of a computer.
- allows components such as sound cards, network cards, and expansion cards to communicate with the CPU and memory
- developed by **Intel** in the early 1990s to replace older bus standards like ISA.
- **PCI provides a high-speed communication pathway between the processor and connected devices inside a computer system.**

Expansion Type Buses....

PCMCIA - Personal Computer Memory Card Industry Association (Also called PC bus)

- is a standard expansion interface developed for portable computers that allows small cards to be inserted to add hardware functions such as networking, storage, or modems.
- The **PCMCIA** developed a standard that allows small, credit-card-sized devices to be inserted into portable computers to expand their capabilities.

Main Purpose

PCMCIA was created to make it easy to add hardware to laptops, such as:

Modems

Network cards (Ethernet or Wi-Fi)

Storage cards

Sound cards

Expansion Type Buses....

Accelerated Graphics Port (AGP)

- a high-speed interface used to connect a graphics card directly to a computer motherboard to improve graphics performance.
- provides a **dedicated pathway** between the graphics card and the system memory, which allows faster transfer of graphics data. This improves video rendering, gaming, and 3D applications.

AGP was introduced by **Intel** in 1997 to replace PCI for graphics processing.

Expansion Type Buses....

PC Card

- a small expansion card used in laptops to add hardware features like networking, storage, or modems, based on the PCMCIA standard.
- allows users to insert a small card into a slot on a laptop to expand its capabilities without opening the computer.

Common Uses of PC Cards

PC Cards were used for:
Network cards (Ethernet or Wi-Fi)
Modems
Extra storage devices
USB expansion
Sound cards

Small Computer System Interface (SCSI)

a standard interface for connecting multiple high-speed peripheral devices to a computer, allowing data transfer between the computer and devices like hard drives, scanners, and printers..

How does a Microprocessor Work?

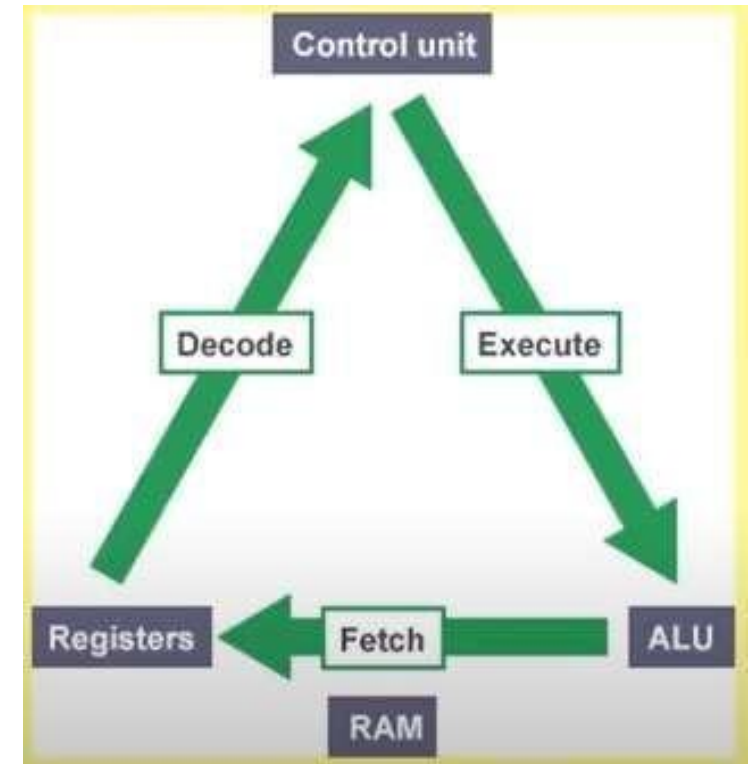
The Microprocessor follows a sequence: **Fetch, Decode, Execute**

Steps in the Execution Cycle:

Fetch: The microprocessor retrieves an instruction from memory.

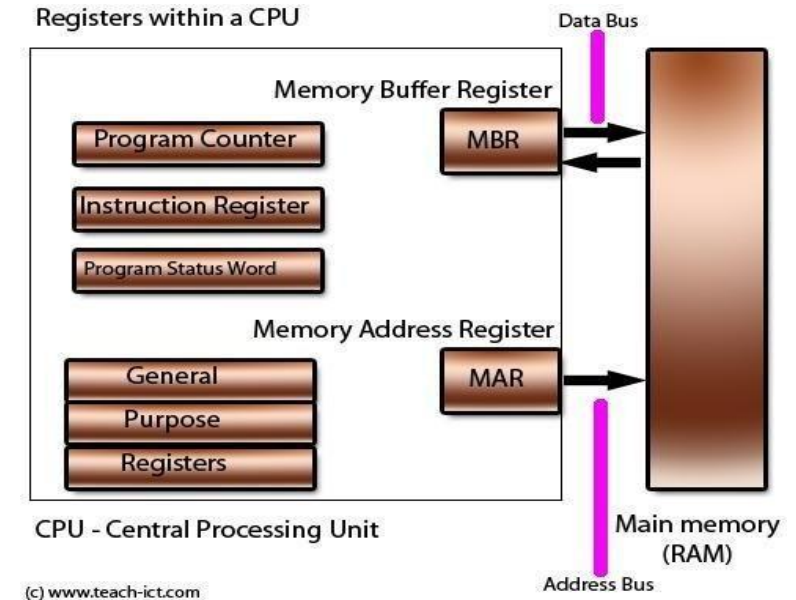
Decode: It interprets the instruction to determine the required operation.

Execute: The Arithmetic Logic Unit (ALU) performs computations.



Fetch

- **Step 1:** The Program Counter copies the address of the next instruction it contains into the Memory Address Register (MAR).
- **Step 2:** The Memory Address Register places the address to be used on to the 'Address Bus'
- **Step 3:** The Memory Address Register triggers a 'read' signal that causes main memory (RAM) to place the instruction being asked for on to the 'Data Bus'
- **Step 4:** The instruction on the data bus is loaded into the Memory Data Register (also called Memory Buffer Register)
- **Step 5:** The Memory Data register copies the instruction into the 'Instruction Register' The Fetch stage is now complete



Decode and Execute

- **Decode:**

- The Control unit examines the instruction in the current instruction register (CIR) and 'decodes' it. This means a special part of Control unit called the 'decode' unit will make the rest of the CPU ready to carry out the instruction.
- For example the instruction might say 'Add'. The decode unit understands what this means and gets the system ready to carry out that instruction.

- **Execute:**

The instruction within the instruction register is carried out (executed) by the ALU. The part that executes instructions is called the 'execute unit'.

Memory-address

What is a Memory Address?

- A **memory address** is simply the **unique number (location)** assigned to each **byte** stored in memory.
- Since memory in computers is **byte-addressable**, each byte has a separate address.
- A program uses these addresses to **store, fetch, or modify data**.

If memory starts at address 00000H, the next byte will be at 00001H, then 00002H, and so on.

If memory size = M bytes, then the required **address size (N bits)** is: $N = \log_2 M$

Examples

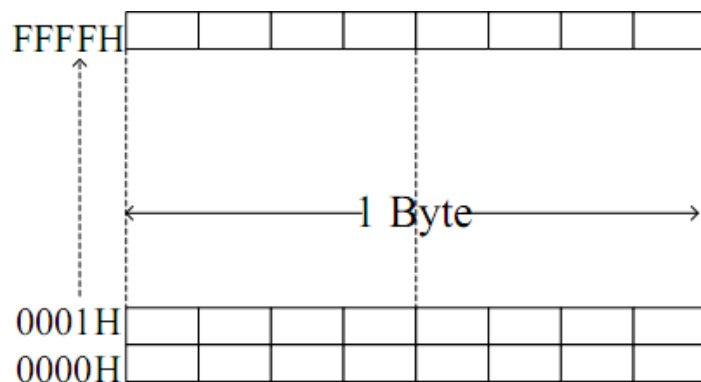
64 KB memory = $64 \times 1024 = 65536 \rightarrow \log_2(65536) = \mathbf{16}$ address lines.

1 MB memory = 1,048,576 bytes $\rightarrow \log_2(1048576) = \mathbf{20}$ address lines.

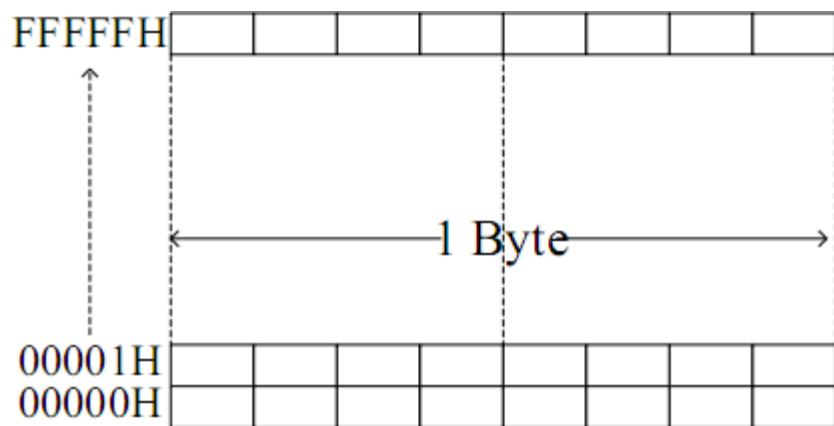
4 GB memory = **32** address lines.

Memory-address

- for addressing 64 KB of memory we need at least 16 bits.



64 Kbyte Memory



1 MByte Memory

- How many bits are necessary to address 1MB memory?**

1MB= 1024Kbyte= 1,048,576 Byte = 2^{20} Byte. So, we need 20 bits to address 1MByte memory. Addresses will be-

0 to 1,048,575 (Decimal)

00000H to FFFFFH (Hex)

Lowest value: 0000 0000 0000 0000 0000 **-(00000H)**

Highest value 1111 1111 1111 1111 1111 **-(FFFFFFH)**

N.B. 8086 has 20-bit address bus. So, it can addresses 1 MB memory

Intel Processor Generations and Architectures

- **Intel 80186 – Foundation for Embedded Systems**
- **Intel 80286 – Enter Protected Mode**
- **Intel 80386 – 32-bit Breakthrough**
- **Intel 80486 – Speed and Integration**
- **Intel Pentium – Superscalar Power**

Intel 80186 – Foundation for Embedded Systems

Introduction Year: 1982

Architecture: 16-bit processor

Clock Speed: 6 MHz to 25 MHz

Address Bus: 20-bit (can address up to 1 MB memory)

Data Bus: 16-bit

Key Features:

- Enhanced instruction set over 8086, allowing faster and more efficient execution
- Integrated system components: interrupt controller, timers, DMA controller,
- Operates in **real mode only** (no protected mode like the 80286)
- Optimized for **embedded systems** rather than desktop PCs

The Intel 80186 was designed as an upgraded version of the 8086, primarily targeting embedded applications. It featured integrated support hardware, which reduced overall system complexity and cost. Although not widely adopted in personal computers, it became popular in embedded devices such as printers, networking hardware, and industrial control systems.

Intel 80286 – Enter Protected Mode

Introduction Year: 1982

Architecture: 16-bit processor with some 24-bit addressing

Clock Speed: 6 MHz to 25 MHz

Address Bus: 24-bit (can address up to 16 MB memory)

Data Bus: 16-bit

Key Features:

- Protected mode support, allowing multitasking and better memory management
- Real mode compatible with 8086 software
- No built-in floating point unit (FPU), but could be used with external 80287 coprocessor
- Improved performance over 8086 with pipelining and better instruction execution

The 80286 was a major step from 8086, introducing protected mode to support advanced OS features and larger memory addressing, making it suitable for business applications and early multitasking systems

Intel 80386 – 32-bit Breakthrough

Introduction Year: 1985

Architecture: 32-bit processor

Clock Speed: 12 MHz to 40 MHz

Address Bus: 32-bit (can address up to 4 GB memory)

Data Bus: 32-bit

Key Features:

- Full 32-bit internal and external architecture
- Supports real mode, protected mode, and virtual 8086 mode
- Built-in Memory Management Unit (MMU)
- Supports multitasking and advanced operating systems
- Optional integrated floating point unit (80387 coprocessor)

The 80386 was Intel's first 32-bit CPU for PCs, vastly increasing memory and processing power, enabling more powerful software and operating systems like Windows NT and UNIX

Intel 80486 – Speed and Integration

Introduction Year: 1989

Architecture: 32-bit processor

Clock Speed: 20 MHz to 100 MHz (depending on model)

Address Bus: 32-bit (4 GB memory addressing)

Data Bus: 32-bit

Key Features:

Main Features:

Integrated FPU (Floating Point Unit):

First Intel CPU to include a built-in math co-processor (in DX versions).

Greatly improved the performance of scientific, engineering, and multimedia tasks.

On-Chip Cache (8 KB):

Included a small but fast **Level 1 (L1) cache**, storing frequently accessed data.

Reduced the need to fetch data from slower main memory.

Intel 80486 – Speed and Integration

Instruction Pipelining:

Could start a new instruction before the previous one finished.

This "pipeline" increased instruction execution speed.

Improved Bus Interface and Memory Handling:

Supported faster memory access.

Worked efficiently with 80386-compatible software and operating systems.

Variants:

80486SX: Cheaper version without FPU.

80486DX: Full version with FPU.

80486DX2/DX4: Used internal clock multiplication for faster performance (e.g., DX2 ran internally at 66 MHz even if external clock was 33 MHz).

Common Uses:

Personal computers (early Windows systems)

Office applications and early multimedia

Academic and business environments needing moderate computing power

Intel Pentium – Superscalar Power

Introduction Year: 1993

Clock Speed: 60 MHz to 3+ GHz (varies by generation)

Data Bus: 64-bit

Address Bus: 32-bit (can access up to 4 GB memory)

The Intel Pentium processor introduced **superscalar design**, enabling parallel execution of instructions and significantly outperforming earlier CPUs like the 486. It was the first mainstream Intel CPU to use a **64-bit data bus** and came with an integrated **FPU**, making it suitable for advanced multimedia, gaming, and productivity software.

Key Features:

Superscalar Design: Executes 2 instructions per clock cycle (U & V pipelines)

Integrated FPU: Faster floating-point operations

On-chip Cache: L1 cache (8 KB instruction + 8 KB data); L2 added later

MMX Technology: Accelerated multimedia processing (Pentium MMX & later)

Branch Prediction & Pipelining: Improves performance